

CE 310 — Week 14 Weekly Quiz

Week: 14 | **Topic:** Bootstrap Confidence Intervals

Due: Closes **11:59 PM Sunday, Nov 29** — brought forward because the Final Written Exam is Monday Nov 30. Taken on D2L

Points: 15 questions $\times$ 1 pt each = 15 pts

Attempts: Two — the higher score counts

Time limit: 25 minutes

Instructions: Answer each question in the format it asks for — most are multiple choice, with one matching, one ordering, one select-all-that-apply and one true/false. This quiz is auto-graded and must be submitted before it closes at the time given above. You may use your notes and readings, but work independently.

This quiz covers the Monday lecture and the Wednesday Python tutorial — bootstrap resampling and percentile confidence intervals, permutation tests, the pairs bootstrap for regression slopes, and the seeded rng habits the notebook depends on. The tutorial's warning boxes are where the most time-consuming mistakes are flagged, so they are worth reading before you start.

Questions

Q1. What distinguishes bootstrap resampling from Monte Carlo simulation (Week 13)?

- A) Bootstrap needs more replicates than Monte Carlo before it converges
 - B) Bootstrap samples from the data itself; Monte Carlo from assumed distributions
 - C) Monte Carlo resamples the observed rows, while the bootstrap assumes a Normal shape
 - D) They are the same method under two names used by different textbooks
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Q2. The mean awarded price for roadway excavation is $\bar{x} = \$28.138/\text{CY}$, with an analytical standard error $\text{SE} = \$1.424/\text{CY}$. **Without using Python**, compute the 95% analytical confidence interval using $\bar{x} \pm 1.96 \times \text{SE}$.

- A) (25.347, 30.929)
 - B) (26.714, 29.562)
 - C) (27.138, 29.138)
 - D) (28.138, 30.929)
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Q3. [Matching] Match each resampling method to what it produces.

Choices:

- (A) Percentile bootstrap CI
- (B) Bootstrap standard error
- (C) Residual bootstrap
- (D) Permutation test
- (E) Pairs bootstrap

Matches:

1. The standard deviation of the resampled statistics
 2. A distribution of what the statistic would look like if the grouping meant nothing
 3. The 2.5th and 97.5th percentiles of the resampled statistics
 4. Regression uncertainty that reflects variability in the predictor as well as the outcome
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Q4. [True/False] A bootstrap standard error of \$1.405/CY and an analytical standard error of \$1.424/CY for the same sample mean is the expected outcome, not a sign that something has gone wrong.

Q5. You bootstrap 1,000 slopes for $\log p \sim \log q$. The bootstrap 95% CI is $[-0.1742, -0.1584]$ and the analytical OLS 95% CI is $[-0.1735, -0.1596]$ — the bootstrap is 13.7% wider. An HC3 robust standard error, computed without any resampling, comes out 1.145× the OLS one. What is the most likely explanation?

- A) The bootstrap carries Monte Carlo error; more replicates would close the gap
 - B) The bootstrap slope must have been computed wrongly somewhere inside that resampling loop
 - C) Residual spread varies with job size, which the OLS standard error assumes away
 - D) OLS intervals are always too narrow, because OLS assumes Normal residuals
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Q6. A permutation test for the winner discount shuffles the LOW labels 1,000 times and measures the gap in mean $\log_{10}$ price each time. None of the 1,000 shuffles produces a gap as large as the observed -0.0706 . What is the permutation p-value?

- A) 0.5, because the null hypothesis puts even odds on either direction
- B) 1.0, because every one of the shuffles exceeded the observed gap
- C) 0.071, read directly off the size of the observed gap itself

- D) 0.0, because zero of the 1,000 shuffles matched or beat the observed
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Q7. [Select all that apply] In which situations does the bootstrap earn its place over an analytical formula?

- (A) The statistic is the median of a small, heavily skewed sample
 - (B) The statistic has no convenient closed-form standard error
 - (C) The statistic is the mean of 10,000 roughly Normal observations
 - (D) The data violate an assumption the analytical formula depends on
 - (E) You want a narrower interval than the analytical method gives
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Q8. A permutation test shuffles the winning-bid labels across the recorded prices and recomputes the gap each time. What does the shuffling accomplish?

- A) It breaks the link between label and price, leaving the prices intact
 - B) It forces the winning and losing groups to hold equal numbers of bids
 - C) It removes the correlation between bids submitted on the same project
 - D) It resamples the prices with replacement, exactly as the bootstrap does
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Q9. A pairs bootstrap (resample whole rows) is preferred over a residual bootstrap (hold predictors fixed, resample residuals) when:

- A) The residuals are Normally distributed around the fitted line
 - B) The predictor was observed rather than set by the analyst
 - C) The sample is large enough that theory can be relied upon
 - D) You would like the resulting confidence interval to be narrower
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Q10. [Ordering] Put the steps of a percentile bootstrap confidence interval in the order you would carry them out.

- (A) Draw a resample of the same size as the original data, with replacement
 - (B) Compute the statistic of interest on that resample and store it
 - (C) Repeat until several thousand values have been stored
 - (D) Take the 2.5th and 97.5th percentiles of the stored values as a 95% interval
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Q11. exc holds all 4,369 bids on the item and won holds the 555 awarded rows. Which one do you bootstrap for the mean **awarded** price?

- A) exc, since a larger sample always yields the more reliable interval

- B) Either one — the mean is unaffected by which of the two you resample
 - C) won, because the awarded rows are the prices the agency actually paid
 - D) exc, then divide the result by the average number of bidders taking part on each job
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Q12. You have 1,000 bootstrap replicates in `boot`. How do you form the 95% percentile confidence interval?

- A) `boot.mean() ± 1.96 * boot.std()`
 - B) `np.percentile(boot, [2.5, 97.5])`
 - C) `np.percentile(boot, [5, 95])`
 - D) `[boot.min(), boot.max()]`
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Q13. A PAIRS bootstrap for a regression slope resamples what?

- A) The x column and the y column independently, each drawn with replacement
 - B) Only the residuals, which are added back to the fitted values while x stays fixed throughout
 - C) The fitted coefficients, drawn from their estimated covariance matrix
 - D) Whole rows, so each draw keeps an (x, y) pair together and the association survives
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Q14. Every bootstrap in the notebook draws from one `rng`, seeded once in the setup cell. You re-run a single Section A cell after finishing Section B. What happens?

- A) Every answer computed after that cell changes, because the stream advanced
 - B) Nothing changes, because the seed is fixed and always replays the same draws
 - C) Only that one cell's answer changes; the cells above and below are unaffected
 - D) The notebook raises an error, since a seeded generator cannot be reused
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Q15. Your permutation test prints `ANSWER_B5 = 0.0` after 1,000 shuffles. What should the written answer say?

- A) The probability of the observed gap arising by chance is exactly zero
- B) The test failed, since a valid p-value can never come out as exactly zero

- C) $p < 1/1000$ — 1,000 shuffles cannot resolve any finer than that
- D) $p = 0.05$, the conventional floor below which p-values are not reported